# Basin hopping on Lennard-Jones clusters with one preset at every size

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**Abstract**

We describe one basin-hopping preset for Lennard-Jones clusters in which the move library, the temperature, the census, and the orbit completion stay fixed at every size. Below 98 atoms the constructor adds a diameter penalty at 0.7 of the largest pair distance, and restart on stall stays off. The novelty is that fixed core, with the depth posterior choosing the move. On the water hexamer the six rows are single draws from CREST 3.0.2, which exposes no seed, while the anneal seed fixes only start.xyz, so those rows are not a success rate: seed 1 lies at $1.35 \times 10^{-7}$ Eh and the other five polished gaps lie below it.

# 1 Introduction

We describe one preset that a reader can run at every cluster size. In contrast, older presets changed with the atom count. A thermodynamic model starts from a local minimum of the potential.[1] The same model treats the global minimum. As the ground state. On the Lennard-Jones pair potential the number of minima grows quickly. With the number of atoms.[2,3] A search is then judged by how many. Evaluations it spends before it reaches a minimum that is already known. Li and Scheraga introduced basin hopping for a rough. Potential-energy surface.[4] Wales and Doye made that.

Monte Carlo walk the working method for atomic clusters.[2] A move proposes a new geometry. A local minimisation returns a minimum. And the Metropolis rule at one temperature accepts or rejects. That minimum. We find the same preset at every size in the campaigns below. The earlier four-arm hexamer campaign is not given counts in this article. The preset measured here draws move arms by Thompson sampling. From a posterior on the depth of the quench. And it abandons a quench that is returning to the occupied basin.[5,6] A stall applies tabu, symmetrisation, the recorded basin trail, and the climb. The constructor installs one move library. Below 98 atoms it installs one two-phase surface, a relative cutoff of. 0.7 with weight 1, and at 98 atoms and above that list stays empty. A caller who wants lattice growth sets it on the configuration.

# 2 The search

## 2.1 One step of basin hopping

A chain stores a minimum $x$ with energy $E(x)$. The move adds a displacement, the local minimiser returns a new minimum. $y$, and the chain accepts $y$. With the Metropolis

probability at temperature $T$,

$$\min\bigl(1, \exp[-(E(y) - E(x))/T]\bigr). \tag{1}$$

$T$ is 0.8 in the reduced units where the pair well. Depth is 1. Wales and Doye use that reduced temperature for Lennard-Jones. Basin hopping.[2] The Lennard-Jones preset stores the value as `TEMPERATURE`. One temperature is used at every size. The hit that scores a run is the first converged evaluation whose. Energy lies below the published minimum by less than 0.001. A miss is a run that reaches its call ceiling without such. An evaluation. The penalised average runtime (PAR-2) is the call count of a hit. And twice the cap for a miss. The hit is read from the trace of converged improvements. The trace stores no stop threshold.

Algorithm 1 lists one chain of the recommended preset.

**Algorithm 1** One chain of the recommended preset.

Draw a non-overlapping start from the seed and quench it.
**while** the call budget remains **do**
    Draw a move from the library. And quench the displaced geometry to a minimum $y$.
    Replace $x$ by $y$. With the probability in equation 1.
    When $y$ is a new basin. Quench the symmetry images of its core.
**end while**

## 2.2 What the preset turns on

The recommended preset starts from the Wales and Doye displacement. And then selects one move library. Thompson sampling draws each move from a normal-gamma posterior. On the depth reached by earlier quenches of that move.[5] Below 98 atoms the constructor sets the surface. List to one diameter penalty. By calling `TwoPhase::relative` with. 0.7 and 1. The cutoff equals 0.7 times the largest pair distance. That distance is read on the structure that enters the relaxation. The penalty weight equals 1. The surface portfolio then draws that penalty or the plain surface, using the same depth posterior. The surface

list stays empty at 98 atoms and above. The relaxation then uses the plain surface. The lattice-regrowth library builds a truncated octahedron. A decahedron, or an icosahedron from a named template. The global minimum changes with size.[2,7] Face-centred cubic. Is the minimum at 38 atoms, icosahedral at 55, and Marks decahedral at. 75. Installing that library only at 38 atoms would search for a structure. The count has already named. The preset therefore keeps the lean burst library at every size, including 38. Doye showed that the Locatelli-Schoen compression. A term that pulls every atom toward the centroid. Is a change of the potential.[8,9] The installed surface is a diameter. Penalty, and its centroid weight stays at 0. Restart on stall stays at its default, off. So a stalled walker is not redrawn from random coordinates. A stall still applies tabu, symmetrisation, the recorded basin trail, and the climb. Orbit completion quenches the images of a symmetric core when a new. Basin appears. A continuous-symmetry projection every ten quenches. And a reseed after five hops on an icosahedral plateau. Follow the defaults Oakley, Johnston and Wales published for the Wales-group basin-hopping program. (GMIN).[10] Those two numbers are applied at every size. Figure 1 stacks the potential, the hop. And the bank that shares the minima. The figure is drawn from the search, and no campaign count enters it.

# 3 Statistical framework

A cooling schedule sets the step. And the temperature of one chain. A population of chains also decides which replicas carry on. The census of packings then decides when the next minimization belongs. To a different packing. We resample replicas on within-epoch ranks, and we close the census. With a Good-Turing estimate of the unseen mass.[12]

## 3.1 Annealed-flow kernel

As the search cools, the replicas follow a sequence of distributions. that weight lower energies more heavily. Del Moral, Doucet, and Jasra write that sequence as the equation below.[13]

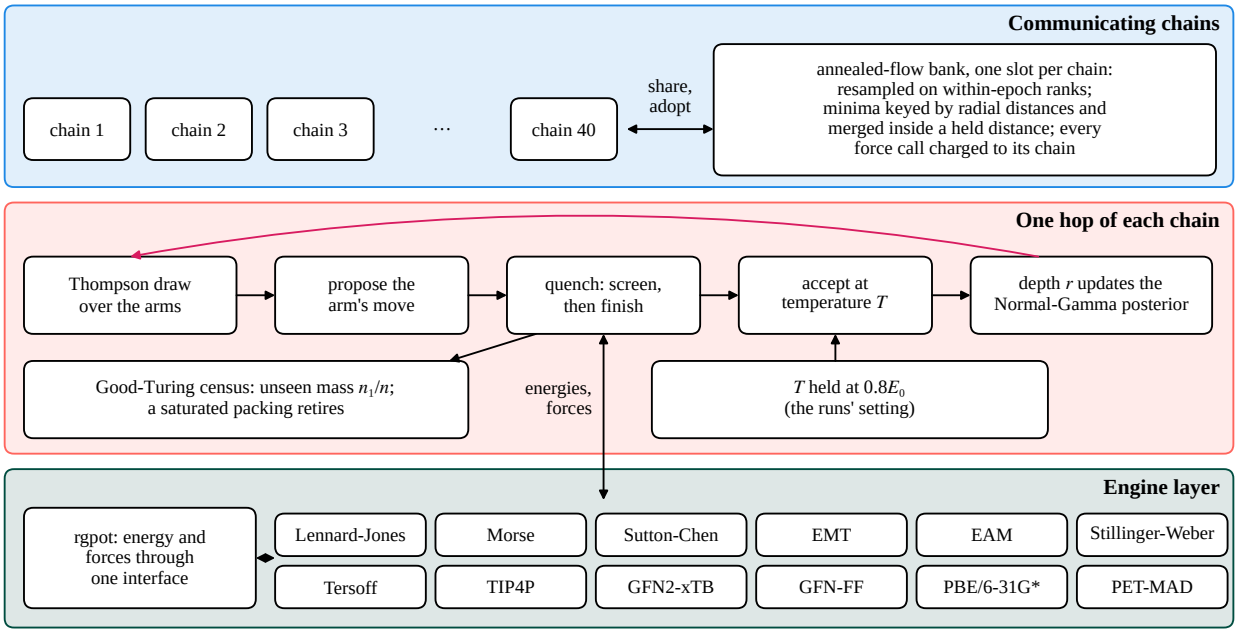



Figure 1: The stack of the search. The chains share minima through the annealed-flow bank. In one hop the Thompson draw picks the arm, the quench is screened. And the acceptance test runs at the held temperature $T = 0.8$. The depth of the quench updates the posterior of the arm that played, and a saturated packing retires. Every potential reaches the hop through rgpot.[11]

$$\eta_t(\mathrm{d}x) \propto \exp(-\beta_t E(x))\,\lambda(\mathrm{d}x), \tag{2}$$

Beta increases along the sequence. They move from one target to the next by reweighting. They then apply a move kernel.

$$\eta_{t+1} = \Psi_{G_t}(\eta_t)\,M_{t+1}. \tag{3}$$

Cluster energies share no scale from one size to the next. The weight is therefore the within-epoch rank. Tied replicas share their average position. Positions run from 0 to $R-1$. A tied block runs from $b$ inclusive to $e$ exclusive. The block has the rank below when more than one replica is present.

$$r = \frac{b + e - 1}{2(R-1)}. \tag{4}$$

A lower energy receives a lower rank. The energy rank $r_E$, the novelty rank $r_N$, the scarcity rank. $r_S$, and the uncertainty rank $r_U$ combine as

$$\log w_i = -r_{E,i} + 0.8\,r_{N,i} + 0.6\,r_{S,i} + 0.4\,r_{U,i}. \tag{5}$$

Offspring go to replicas that are low in energy, rare as packings. And uncertain as basins.

## 3.2 Occupancy and the temperature of the hop

Those packings and basins set the occupancy of the hop. The Good-Turing estimate of the unseen mass closes that census.[12] A saturated packing. Then retires. The next minimization is spent on a packing the census has barely seen. The temperature of the recommended hop stays at the Wales and Doye value 0.8 in reduced units.[2] A caller who wants a schedule that tracks the entropy. Of the occupied states sets it on the configuration. The preset itself keeps one temperature at every size.

# 4 Communicating chains

Many searches already run many chains. We call the communication correct when five conditions hold. Each chain draws its next move from a posterior on the depth. That move reaches. A packing retires on the Good-Turing estimate of its unseen mass. Minima are shared under one radial merge distance. Every force evaluation is charged to the chain that spent it. So chains are compared at equal budgets. The receiver can act on a shared minimum by quenching it. And entering it in the acceptance test.

Parallel tempering (PT) exchanges states along a ladder of temperatures.[14,15] Population annealing resamples a population by its Boltzmann weight.[16,17] Conformational space annealing (CSA) keeps a bank of minima.[18] CSA also anneals. The distance cutoff. Deaven and Ho mate stored geometries by cut and splice.[19] The Birmingham pool algorithms do the same.[20] Basin hopping (BH) and minima hopping (MH) run a single walker.[2,6] The conformer-rotamer ensemble sampling. Tool (CREST) samples a molecular ensemble by metadynamics.[21] The bank in this section adds. The depth posterior and the unseen-mass census to that list.

# 5 Information theory in the hop

A move that has stopped finding depth can still take. Over the next trial. The recommended hop guards against that takeover. With a posterior on the depth each move reaches.

## 5.1 Depth posterior

On a deep cluster few of the available moves improve the incumbent. We draw the next arm by Thompson sampling from a normal-gamma posterior. On the depth earlier quenches of that arm reached.[5] The depth that updates the posterior is the drop. From the energy before the quench to the energy after it. The same acceptance test meets every arm. With the well-tempered bias at zero. That test is equation 1.[22] Figure 2 draws. The arms and the

capped burst length. The figure is drawn from the search, and no campaign count enters it.

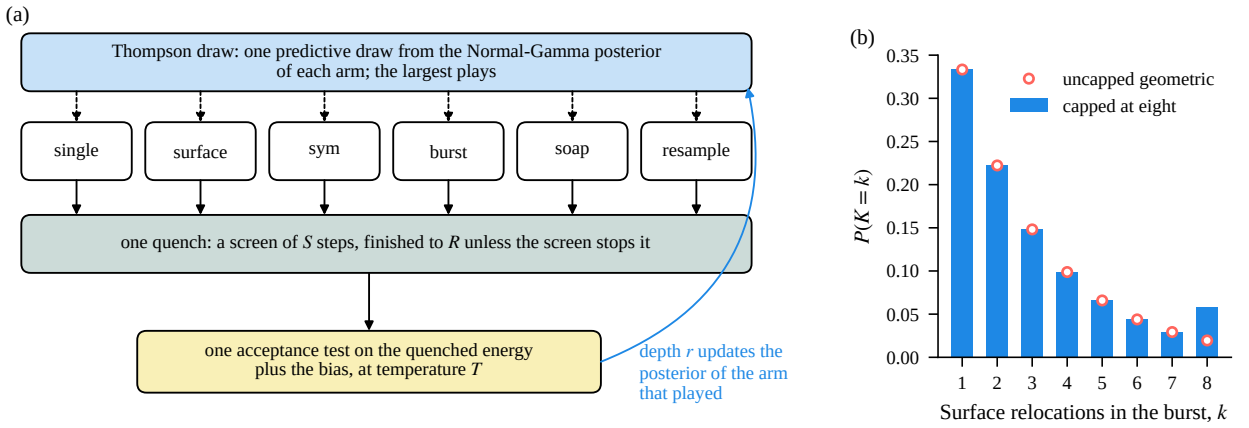



Figure 2: The figure shows the arms of the recommended hop. And the law of its burst. The Thompson draw picks one arm. Whichever arm plays, its proposal is quenched once. That proposal meets one acceptance test. The depth then updates the posterior. The burst length is capped at eight relocations.

## 5.2 Symmetrisation

The burst drawn above is one arm of the hop. After that quench. The search projects the structure toward a more symmetric neighbour. A global minimum is often more symmetric than the minima around it. On $LJ_{75}$ the global minimum is a Marks decahedron.[7,23] Most. Low minima around that decahedron are icosahedral. Oakley, Johnston, and Wales symmetrise the structure in GMIN.[10] One route completes the orbits of a symmetric core. Another route uses a continuous symmetry measure. The preset applies that projection every ten quenches at every size.

# 6 Results

## 6.1 The structures

Figure 3 draws the archived structures of the benchmark. The Lennard-Jones geometries are the reference minima of the Cambridge. Cluster Database, and the TIP4P geometries

are the minima of Wales and. Hodges.[24–26] The hexamers are minima of the hop at the GFN2-xTB tight-binding model.[27] The Supporting Information. Draws the full structure plates.

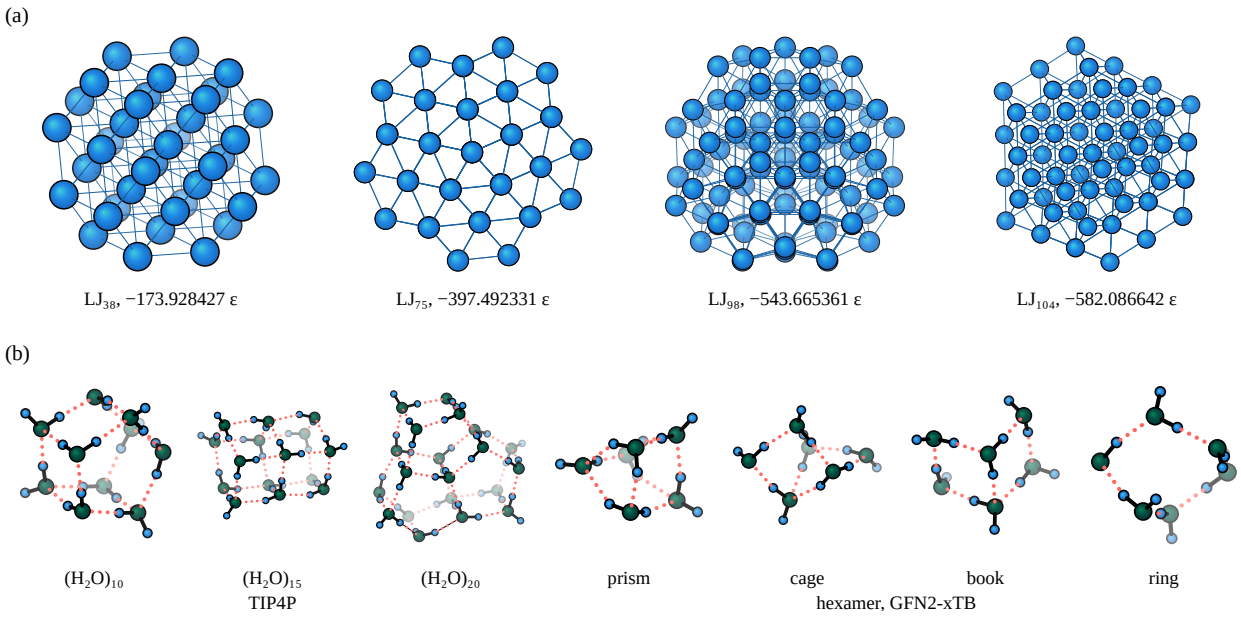



Figure 3: Structures of the benchmark. (a) The global minima of $LJ_{38}$, $LJ_{75}$, $LJ_{98}$ and $LJ_{104}$. (b) The TIP4P minima of 10, 15 and 20 molecules. And the lowest GFN2-xTB hexamer of each class.

The $LJ_{75}$ seed sweep runs serially at a cap of. $4 \times 10^6$ force calls. No hit count for those seeds is in the sealed record. Figure 4 draws the encounter curves of the earlier. Sealed $LJ_{75}$ campaign. The old-method baseline on $LJ_{104}$ is measure v4.14. Its six receipts, seeds 2000 through 2005, record 0 of 6 hits, and the best of those receipts is −581.80180238 on seed 2005, against the reference −582.086642.

## 6.2 Rates and the water hexamer

Figure 5 maps the minima one seed of basin hopping and of the recommended hop accepted at $LJ_{38}$, $LJ_{75}$ and. $LJ_{98}$. At $LJ_{38}$ the recommended hop reached the truncated octahedron. The $LJ_{75}$ minima on that map reach the icosahedral competitor at. −396.282249, which is 1.210 above the decahedral global minimum. At −397.492331. The $LJ_{98}$ minima stop 0.022

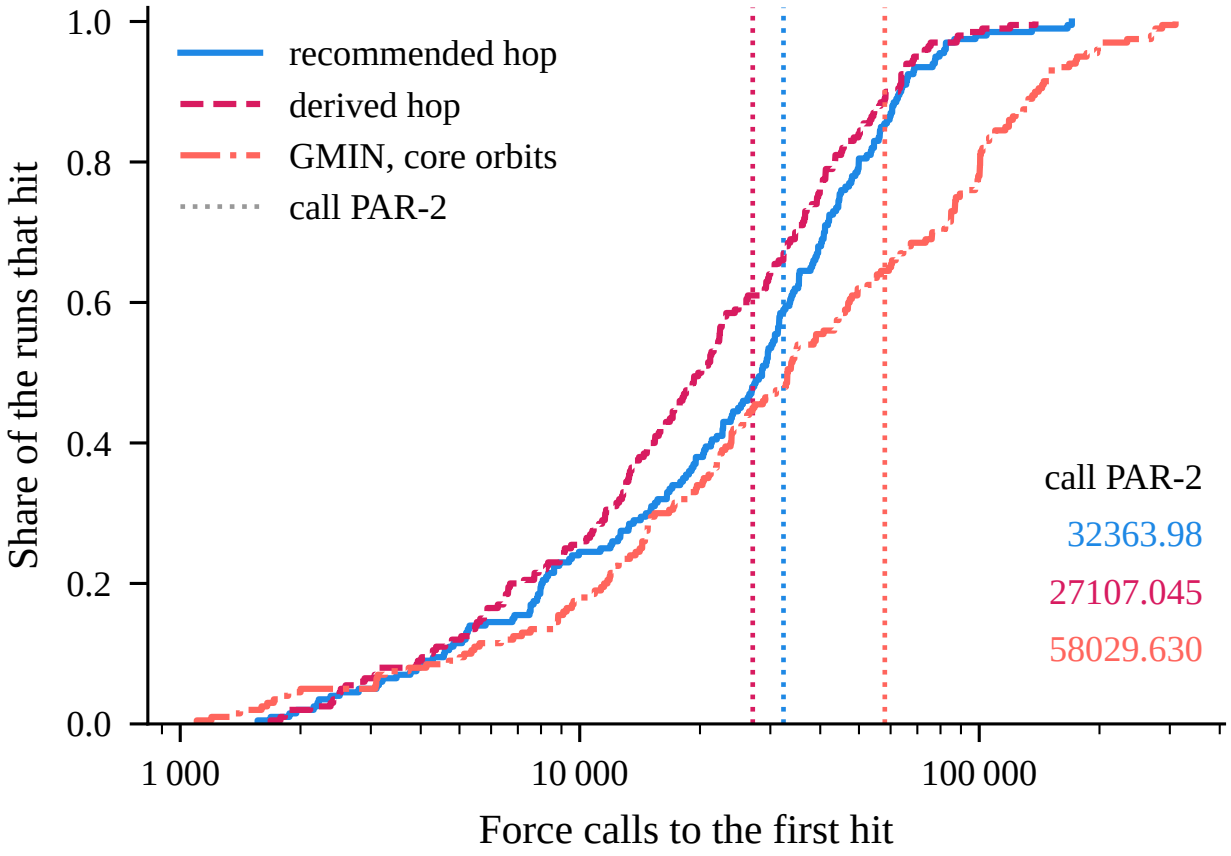



Figure 4: Encounter curves of main's earlier sealed $LJ_{75}$ campaign. The panel prints that run's PAR-2 values. The run used an empty surface list and restart on stall. The arms are the recommended hop, the derived hop. And the Wales-group basin-hopping program (GMIN). The plot covers that sealed run.

above the tetrahedral global. Minimum at −543.665361. Figure 6 joins the $LJ_{38}$ minima where. They first connect among their nearest neighbours. In the coordination descriptor. A search records no transition states. So no join of the tree is a barrier. Figure 7 shows a landfold panel for each sealed Lennard-Jones size and for the TIP4P hexamer.

## 6.3 Earlier four-arm hexamer campaign

The earlier four-arm hexamer campaign is not reported here. Its raw seed logs are not in the record. The hit counts and the McNemar tests are omitted.[21,27] The default arm is the hop. The budget-fair polish is an opt-in mode that charges an external polish. The third arm is CREST. And the fourth runs that polish after CREST. A run would hit when the iterative rotations and assignments alignment (IRA) matches a structure to a Malloum prism, cage, book, or cyclic chair reoptimised at GFN2-xTB.[28–30]

## 6.4 Water hexamer against CREST

CREST here is conda-forge crest 3.0.2. Its metadynamics is stochastic for a fixed start geometry. This build exposes no seed. The anneal seed fixes only start.xyz. CREST's cost is the energy and gradient call count. The matched comparison. And the multi-start statistics are not in this article. A row succeeds when the run exits cleanly. The polished energy is finite. And the maximum force is at or below 0.001 Eh/bohr. Every final is polished with xtb at GFN2, the vtight setting, accuracy. 0.01, and an iteration limit of 500. The polish uses xtb's vtight setting.

Table 1 lists two CREST counters for each seed. The old counter sums the dynamics steps and the optimization steps. CREST prints a second total, "Total number of energy+grad calls". The old counter undercounted CREST on every seed.

The 4000-call arm spends 4000 evaluations in one setting. The driver-count budget is the anneal arm whose cap equals each seed's. Old counter. The driver-count budget runs from 32112 to 50063 evaluations. CREST's energy and gradient totals run from 142110 to

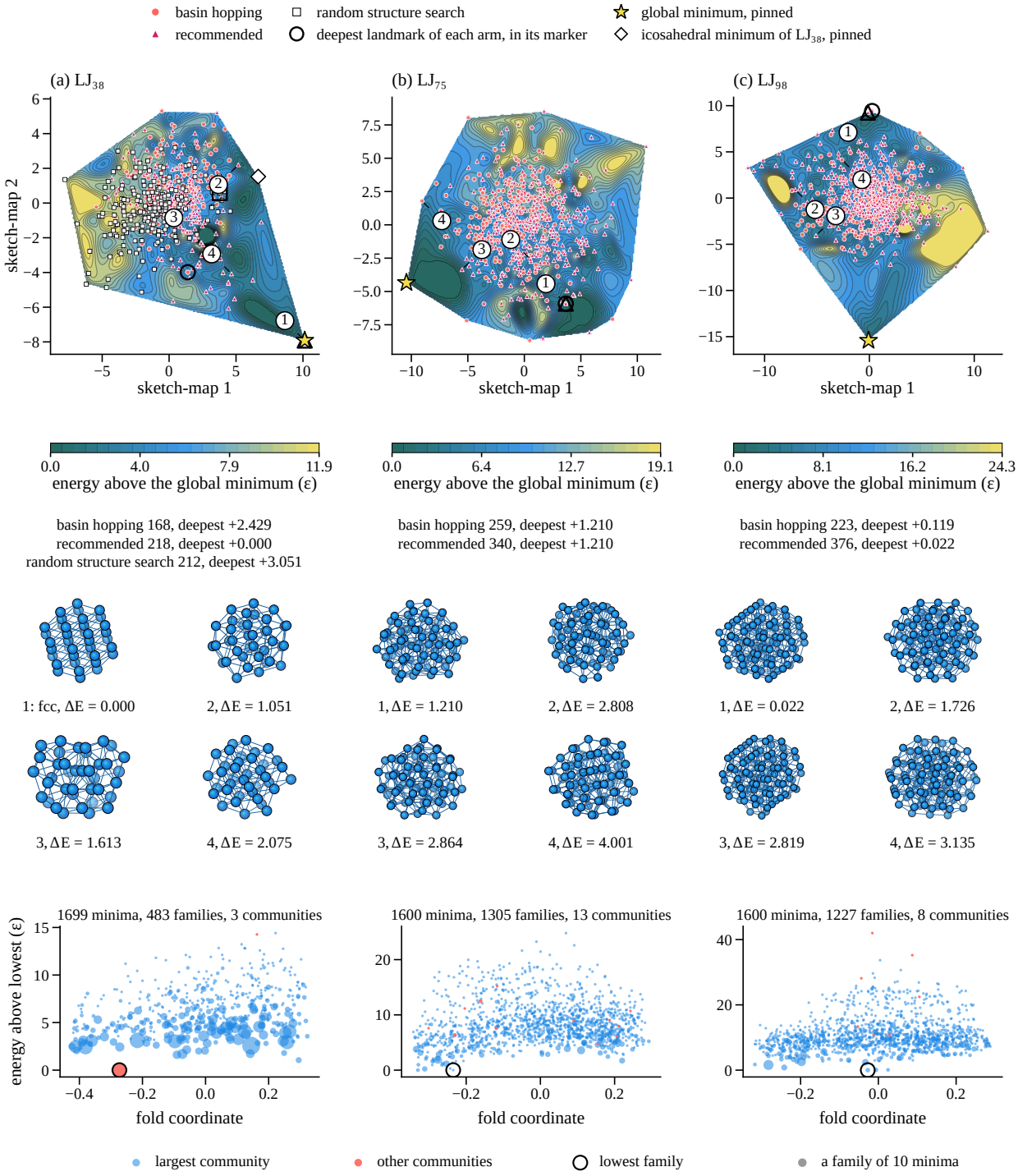



Figure 5: Maps of the minima one seed accepted at $LJ_{38}$, $LJ_{75}$, and $LJ_{98}$. Top, the sketch-map. Middle, four deep landmarks. Bottom, the landfold map of the same minima. Sky in the largest community and coral in the others.

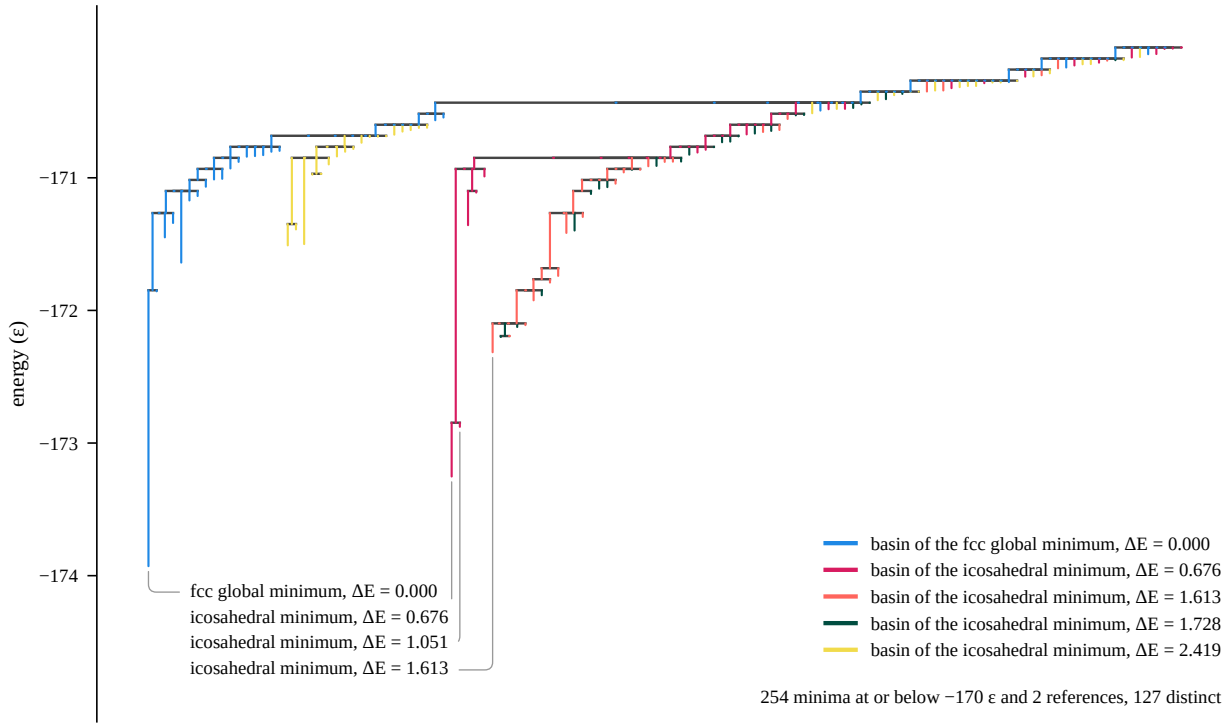



Figure 6: Approximate disconnectivity graph of $LJ_{38}$. Nearby minima in the coordination descriptor are merged. The stems are coloured by five spectral basins. No join is a barrier.

Table 1: CREST effort on the water hexamer. The old counter is dynamics steps plus optimization steps. The energy and gradient column is CREST's own summary line.

| Seed | Old counter | Energy+gradient |
|---|---|---|
| 0 | 32112 | 151074 |
| 1 | 50026 | 191132 |
| 2 | 32155 | 142110 |
| 3 | 46062 | 172468 |
| 4 | 50063 | 157643 |
| 5 | 32170 | 209931 |

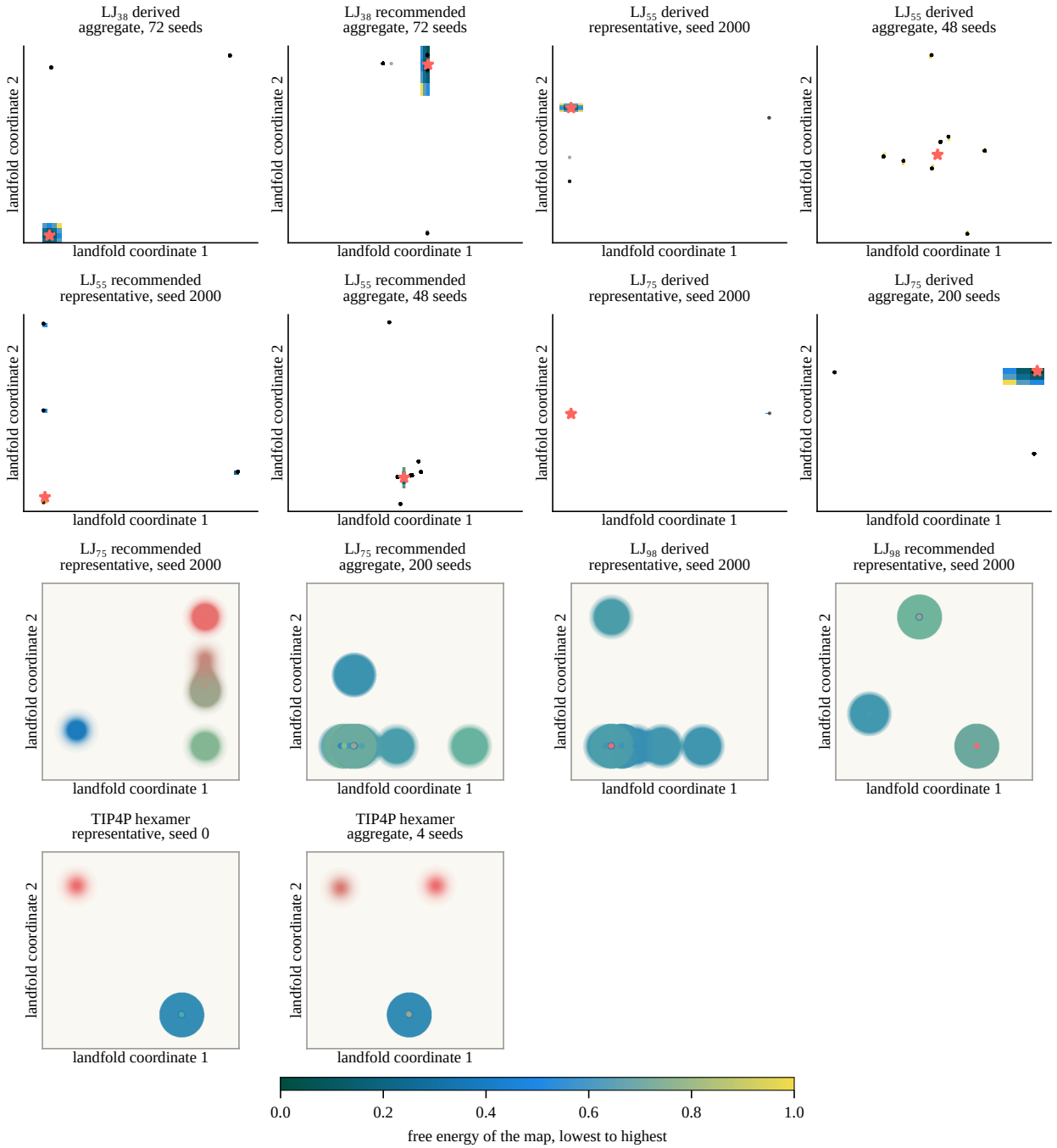



Figure 7: Landfold maps of the sealed Lennard-Jones and TIP4P hexamer runs. The Lennard-Jones panels use main's preset, including an $LJ_{38}$. derived panel and an $LJ_{75}$ aggregate. Each panel is one sealed map. Coloured from the lowest free energy to the highest. With the minimum marked.

209931. The comparison is the 4000-call arm against those energy. And gradient calls. The driver-count budget is not equal effort. Because the old counter undercounted CREST.

Table 2 gives the polished gap on each seed. The gap is the anneal energy minus the CREST energy. A negative value means the anneal energy sits lower. Five seeds lie below seed 1 on both gap ledgers. Seed 1 exceeds that line, at $1.3509 \times 10^{-7}$ Eh on the. 4000-call ledger. And $1.3505 \times 10^{-7}$ Eh on the driver-count ledger. The label-free root-mean-square deviation of each polished pair. Is at most 0.00212 Å.

Table 2: Polished gap and label-free deviation on the water hexamer. The gap is anneal minus CREST, in hartree. The deviation is the label-free root-mean-square deviation. In angstroms.

| Seed | Gap, 4000-call ledger | Gap, driver-count ledger | Deviation, 4000-call ledger | Deviation, driver-count ledger |
|---|---|---|---|---|
| 0 | $-1.1214 \times 10^{-8}$ | $-1.1210 \times 10^{-8}$ | 0.001004 | 0.000987 |
| 1 | $-1.3509 \times 10^{-7}$ | $-1.3505 \times 10^{-7}$ | 0.002023 | 0.001964 |
| 2 | $-2.5460 \times 10^{-9}$ | $-2.5850 \times 10^{-9}$ | 0.000532 | 0.000477 |
| 3 | $-4.1510 \times 10^{-9}$ | $-4.1430 \times 10^{-9}$ | 0.000322 | 0.000308 |
| 4 | $-6.8980 \times 10^{-8}$ | $-6.9022 \times 10^{-8}$ | 0.002116 | 0.002068 |
| 5 | $-2.1900 \times 10^{-9}$ | $-2.2000 \times 10^{-9}$ | 0.000157 | 0.000134 |

The logs name the processor Intel Xeon Processor, family 6, model 207. Each CREST run selected one thread. The hexamer wall-time medians, from `logs/hexamer-walls.txt`, are 544.0 s for CREST, 18.8 s for the 4000-call arm. And 195.5 s for the driver-count budget.

The CREST polish converged in one iteration on all 18 finals. No CREST structure dissociated.

The water 16-mer count is separate open work. Apart from the hexamer campaign. This comparison leaves out the Atomic Simulation Environment. The atomistic global optimization framework (AGOX). And the Wales-group basin-hopping program (GMIN).

# 7 Conclusions

The earlier four-arm hexamer campaign is not given hit counts here. Five of the six hexamer seeds lie below seed 1 on both. Gap ledgers. Seed 1 lies above that line. No hit count for the unfinished $LJ_{75}$ seed sweep is in the sealed record.

## Author contributions

Author contributions follow the CRediT taxonomy. Rohit Goswami contributed conceptualization, methodology, software, formal analysis, and the original draft. Ruhila Goswami contributed software and visualization. Amrita Goswami contributed methodology and edited the draft. Moritz Sallermann contributed software and methodology. And he edited the draft. Sonaly Goswami contributed validation and checked the draft. Debabrata Goswami contributed supervision and oversaw the draft.

## Supporting Information

Board constants, reference-structure plates, sketch-map panels, and landfold panels of the Lennard-Jones maps (PDF)

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## Notes

The authors declare no competing financial interest.

## Data Availability Statement

The data supporting the conclusions are in this article, the Supporting Information, and the public repositories named here. The source is the repository `https://github.com/HaoZeke/anneal`. under the Massachusetts Institute of Technology (MIT) licence. Documentation is at `https://anneal.rgoswami.me`. The archive of the software line is the Zenodo concept record. `https://doi.org/10.5281/zenodo.10672746`. The Zenodo digital object identifier (DOI) for this submission is not assigned. The crate records version 0.12.0. The release tag for version 0.12.0 is not assigned. The copper numbers come from the slab runs. Commit aef5cf0b2068c4988509066a881eb68682ad9290 produced them. The rerun best energies run from -415.971521 eV to -415.971161 eV. The hit tolerance is 0.001 eV. The script is `scripts/elja_hq_sci_molslab_bank.sh`. The lines are 118 and 119. TARGET_TOL there is 1e-3. The per-seed gaps run from 8e-6 eV to 3.68e-4 eV above -415.971529 eV. No seed's recorded best equals the reference. These are not exact hits. Eight seeds lie inside the 0.001 eV tolerance. The `set_relative_0.01` line records the 0.01 eV rate. That line is a separate figure. The library pins rgpot 25c759da82eececa8897e4634046fab27409c81a. The library sha256 follows.

d399ef1f675e97c1f5e401a987d6a9df1bc30b3a37ba6586044b66d44be8399d.

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